Question ID: 3934dd0a

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear equations in one variable and systems of equations in two variables	Medium

Question

Which quadratic equation has exactly one distinct real solution?

Answer

A. $(x+15)^2=0$

B. $(x+15)^2=-45$

C. $(x+15)^2=45$

D. $(x+15)^2=135$

Correct Answer: A

Rationale

Choice A is correct. Taking the square root of both sides of $(x+15)^2=0$ yields $x+15=0$. Subtracting 15 from both sides of this equation yields $x=-15$. Therefore, this equation has exactly one distinct real solution.

Choice B is incorrect. This equation has no distinct real solutions, not exactly one distinct real solution.

Choice C is incorrect. This equation has exactly two distinct real solutions, not exactly one distinct real solution.

Choice D is incorrect. This equation has exactly two distinct real solutions, not exactly one distinct real solution.